

DATA STRUCTURE

LAB PROGRAMS

7. Write a C program to find Fibonacci series using Recursion.

```
#include <stdio.h>

int fibonacci(int n)
{
    if(n == 0)
        return 0;
    else if(n == 1)
        return 1;
    else
        return fibonacci(n - 1) + fibonacci(n - 2);
}

int main()
{
    int n, i;
    printf("Enter number of terms: ");
    scanf("%d", &n);
    printf("Fibonacci Series: ");
    for(i = 0; i < n; i++)
    {
        printf("%d ", fibonacci(i));
    }
    return 0;
}
```

Output

```
Enter number of terms: 9
```

```
Fibonacci Series: 0 1 1 2 3 5 8 13 21
```

```
=== Code Execution Successful ===
```